

DecodeLabs Data Analytics Project

SQL Insights Report

Title Page

Project Title: SQL Data Analysis Project

Internship Program: DecodeLabs Data Analytics Internship

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Tools Used:

- MySQL Workbench
 - SQL
 - Microsoft Excel
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Executive Summary

This project focuses on analyzing transactional business data using SQL queries and relational database techniques. Using SQL aggregation functions, filtering conditions, grouping operations, and sorting methods, raw transactional data was transformed into meaningful business intelligence.

The project successfully identified product performance trends, customer acquisition patterns, revenue distribution, and purchasing behaviors using structured SQL queries.

Global Performance Indicators (KPIs)

Financial Performance Overview

Total Orders Processed:

1200 Orders

Total Revenue Generated:

\$1,264,761.96

Average Order Value (AOV):

\$1,053.90

KPI Observation

The business demonstrates strong revenue generation with an average order value exceeding \$1,000, suggesting customers frequently purchase relatively high-value products rather than low-cost items.

Key Insights & Analytical Trends

A. Product Performance Analysis

Best Selling Product by Quantity

Chair generated the highest total quantity sold:

- Quantity Sold: 562 Units
- Revenue Generated: \$195,620

This indicates strong market demand for chair products.

Highest Revenue Generating Product

Chair also generated the highest revenue:

- Revenue: \$195,620

Closely followed by:

- Printer: \$195,612
- Laptop: \$192,126

This suggests that multiple product categories contribute significantly to revenue rather than relying on a single product.

Lowest Performing Product Category

Phone generated the lowest revenue among product categories:

- Revenue: \$151,722

This may indicate lower demand or lower pricing compared to other categories.

B. Customer Acquisition & Referral Analysis

Strongest Customer Acquisition Channel

Instagram generated the highest customer volume and revenue:

- Orders Generated: 259
- Revenue Generated: \$275,285

This demonstrates that social media marketing plays a critical role in customer acquisition.

Second Strongest Channel

Email marketing produced:

- Orders: 250
- Revenue: \$261,808

Email campaigns appear to remain effective despite slightly lower performance than Instagram.

Lowest Performing Channel

Referral traffic produced:

- Orders: 222
- Revenue: \$226,815

Although referral sources generated the lowest revenue among channels, they still contribute significantly to total business performance.

C. Revenue Distribution Insights

Revenue is distributed relatively evenly across products and acquisition channels.

This suggests:

- Reduced dependency on a single revenue source
 - Better risk distribution
 - More stable business operations
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Business Recommendations

1. Increase Social Media Investment

Instagram consistently generated the strongest acquisition performance.

Recommendation:

Increase marketing budget allocation toward Instagram campaigns to maximize ROI.

2. Develop Product Bundling Strategies

Since Chairs, Printers, and Laptops are the strongest performing products, bundle complementary products together to increase order value.

Examples:

- Laptop + Monitor Bundle
 - Desk + Chair Bundle
 - Printer + Tablet Bundle
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3. Strengthen Referral Programs

Referral channels generated the lowest results among acquisition channels.

Recommendation:

Introduce referral rewards programs to improve customer participation.

4. Focus on Underperforming Products

Phone products produced the lowest revenue.

Recommendation:

Analyze pricing strategy, promotions, or customer demand to improve product performance.

Conclusion

This SQL Data Analysis project successfully transformed transactional data into structured business intelligence using SQL query techniques.

By utilizing SQL functions such as:

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- COUNT()
- SUM()
- AVG()

the project identified critical patterns across products, customers, and revenue generation.

The findings indicate that strong product diversification and social media marketing are the primary drivers of business performance. Overall, this project demonstrates practical SQL skills and highlights the importance of data-driven decision-making in business environments.